



DR. ABDULAZIZ ATTAALLAH

Associate Professor of Computer Science

Software Engineering, AI & Digital Transformation

King Abdulaziz University, Jeddah, Saudi Arabia

Email: aattallah@kau.edu.sa | ORCID: 0000-0001-5363-040X

Google Scholar: scholar.google.com/citations?user=scCDFpcAAAAJ

Mobile: (+966) 556891916

EXECUTIVE PROFILE

Associate Professor of Computer Science at King Abdulaziz University with a senior academic profile combining software engineering research, academic governance, digital transformation leadership, enterprise systems orientation, and technology strategy. Research expertise spans cybersecurity, artificial intelligence, blockchain applications, usable security, IoT/IoHT, healthcare informatics, big data security, and decision-support systems. Experienced in teaching, curriculum development, academic advising, research supervision, and leading university digital platforms.

STRATEGIC POSITIONING

Academic Leadership	Digital Transformation	Software Engineering	Research Impact
Research supervision, academic governance, student affairs, committee service	University knowledge portal leadership, technology modernization, strategic platform governance	Requirements, modeling, architecture, design, quality, and secure software systems	Peer-reviewed publications across indexed journals in cybersecurity, AI, blockchain, and healthcare tech

EXECUTIVE ORIENTATION

Technology Leadership Bridges software engineering, digital transformation, platform governance, and academic technology strategy.	Academic Impact Combines teaching, research supervision, curriculum development, academic advising, and quality assurance.	Research Direction Focused on secure intelligent systems, healthcare technologies, blockchain-enabled security, IoT/IoHT, and decision-support models.
--	--	--

CORE COMPETENCIES

Enterprise & Solution Architecture Aligning technology initiatives with organizational objectives, capability mapping, and end-to-end solution architecture.	Digital Transformation & Technology Strategy Strategic planning, technology modernization, innovation management, and transformation leadership.
Requirements Engineering & Systems Analysis Requirements elicitation, stakeholder engagement, business process analysis, and system modeling.	Software Architecture & Distributed Systems Software design, microservices architecture, distributed systems, and enterprise application integration.
Project, Risk & Technology Governance Project planning, risk management, governance frameworks, and execution oversight.	Artificial Intelligence & Emerging Technologies AI applications, intelligent systems, data-driven decision support, and technology innovation.
Cybersecurity & Secure Software Engineering Security assessment, secure software design, blockchain-enabled security solutions, and risk mitigation.	Research Leadership & Academic Quality Assurance Research supervision, curriculum development, outcome-based education, and accreditation activities.

EDUCATION

Degree	Institution	Year
Ph.D. in Software Engineering	University of North Dakota, USA	2017
Master's Degree	Gannon University, USA	2013
Bachelor's Degree in Computer Science	King Abdulaziz University, Saudi Arabia	2007

LEADERSHIP & ACADEMIC APPOINTMENTS

Role	Institution / Unit	Period
Associate Professor, Computer Science Department	King Abdulaziz University	2024 - Present
Member, Student and Alumni Affairs Committee	Faculty of Computing and Information Technology, KAU	2022 - Present
General Supervisor, Center of the University Knowledge Portal	Deanship of Information Technology, KAU	2020 - 2021
Vice General Supervisor, Center of the University Knowledge Portal	King Abdulaziz University	2019 - 2020
Computer Science Department Council Secretary	Faculty of Computing and Information Technology, KAU	2018 - 2019
Assistant Professor, Computer Science Department	Faculty of Computing and Information Technology, KAU	2017 - 2024
Teaching Assistant	Deanship of Information Technology, KAU	2008 - 2017

RESEARCH IDENTITY

Research Themes	Software Engineering • Artificial Intelligence • Cybersecurity • Blockchain Systems • Internet of Things • Internet of Healthcare Things • Healthcare Informatics • Human-Computer Interaction • Big Data Security • Accessibility Engineering • Software Architecture • Decision-Making Models • Fuzzy Logic
-----------------	---

SELECTED ACADEMIC & PROFESSIONAL CONTRIBUTIONS

- Led and contributed to university-level digital platform initiatives through the University Knowledge Portal Center.
- Participated in academic governance through department council service, student and alumni affairs, academic advising, and committee membership.
- Developed and delivered courses across software engineering, artificial intelligence, programming, object-oriented analysis and design, senior projects, and graduate research.
- Supervised and supported undergraduate and graduate research activities in computing, software systems, and intelligent technologies.
- Published peer-reviewed work across cybersecurity, secure software systems, blockchain-enabled applications, healthcare technologies, fuzzy decision-making, and big data security.

TEACHING PORTFOLIO

Software Engineering CPCS-351 Software Engineering I; CPCS-353 Software Engineering Practices; CPCS-454 Object-Oriented Analysis and Design	Programming & AI CPCS-202 Programming I; CPCS-203 Programming II; CPCS-206 Introduction to Programming and Visual Basic; CPCS-336 Introduction to Artificial Intelligence
Projects & Research CPCS-498 Senior Project I; CPCS-499 Senior Project II; CPCS-699 Master's Research	Professional Training CPCS-323 Summer Workplace Training

CERTIFICATIONS, TRAINING & PROFESSIONAL DEVELOPMENT

Leadership Management International • Healthcare of Things • Usability Testing • Software Architecture • Mobile Applications • Usable Security • Accessibility • Expert Evaluation • Human-Computer Interaction • Internet of Things • UI Computational Metrics • Fuzzy Logic

TECHNOLOGIES & PLATFORMS

Java • UML / PlantUML • Mobile Application Development • Flutter • Blackboard Ultra • Software Modeling & Design Tools • Learning Management Systems (LMS) • AI-Assisted Development Technologies

PROFESSIONAL AFFILIATIONS & SERVICE

- Member, Upsilon Pi Epsilon - International Honor Society for the Computing and Information Disciplines, North Dakota State University (2014).
- Academic Advisor, Computer Science Department, King Abdulaziz University (2017-Present).
- Former President, Saudi Student Association, North Dakota State University (2015-2017).
- Software Developer, Deanship of Information Technology, King Abdulaziz University (2008-2009).
- Technician, Deanship of Graduate Studies, King Abdulaziz University (2007-2008).

PUBLICATIONS

- 1. Security Test Case Prioritization Through Ant Colony Optimization Algorithm.** Computer Systems Science and Engineering, vol. 47, pp. 3165-3195, 2023. DOI: 10.32604/csse.2023.040259
- 2. Enhancing Healthcare Management During Covid-19: A Patient-Centric Architectural Framework Enabled by Hyperledger Fabric Blockchain.** Information, vol. 14, pp. 425, 2023. DOI: 10.3390/info14080425
- 3. A Trailblazing Framework of Security Assessment for Traffic Data Management.** Intelligent Automation & Soft Computing, vol. 37, pp. 1853-1875, 2023. DOI: 10.32604/iasc.2023.039761
- 4. A Fuzzy TOPSIS-Based Approach for Comprehensive Evaluation of Bio-Medical Waste Management: Advancing Sustainability and Decision-Making.** Sustainability, vol. 15, pp. 12565, 2023. DOI: 10.3390/su151612565
- 5. Fuzzy-Based Unified Decision-Making Technique to Evaluate Security Risks: A Healthcare Perspective.** Mathematics, vol. 11, pp. 2554, 2023. DOI: 10.3390/math11112554
- 6. A Security Testing Mechanism for Detecting Attacks in Distributed Software Applications Using Blockchain.** PLOS ONE, vol. 18, pp. e0280038, 2023. DOI: 10.1371/journal.pone.0280038
- 7. Analyzing the Big Data Security Through a Unified Decision-Making Approach.** Intelligent Automation & Soft Computing, vol. 32, pp. 1071-1088, 2022. DOI: 10.32604/iasc.2022.022569
- 8. Estimating Usable-Security Through Hesitant Fuzzy Linguistic Term Sets Based Technique.** Computers, Materials & Continua, vol. 70, pp. 5683-5705, 2021. DOI: 10.32604/cmc.2022.021643
- 9. Device Security Assessment of Internet of Healthcare Things.** Intelligent Automation & Soft Computing, vol. 27, pp. 593-603, 2021. DOI: 10.32604/iasc.2021.015092
- 10. Prediction of Covid-19 Pandemic Spread in Kingdom of Saudi Arabia.** Computer Systems Science and Engineering, vol. 37, pp. 313-329, 2021. DOI: 10.32604/csse.2021.014933
- 11. Estimating the Impact of Covid-19 Pandemic on the Research Community in the Kingdom of Saudi Arabia.** Computer Modeling in Engineering & Sciences, vol. 126, pp. 419-436, 2021. DOI: 10.32604/cmesc.2021.014263
- 12. Standalone Behaviour-Based Attack Detection Techniques for Distributed Software Systems via Blockchain.** Applied Sciences, vol. 11, pp. 5685, 2021. DOI: 10.3390/app11125685
- 13. A Fuzzy Multi-Objective Covering-Based Security Quantification Model for Mitigating Risk of Web Based Medical Image Processing System.** International Journal of Advanced Computer Science and Applications, 2020.
- 14. Attribute Based Honey Encryption Algorithm for Securing Big Data: Hadoop Distributed File System Perspective.** PeerJ Computer Science, vol. 6, pp. e259, 2020. DOI: 10.7717/peerj-cs.259
- 15. Managing Security-Risks for Improving Security-Durability of Institutional Web-Applications: Design Perspective.** Computers, Materials & Continua, vol. 66, pp. 1849-1865, 2020. DOI: 10.32604/cmc.2020.013854
- 16. A Hybrid Fuzzy Rule-Based Multi-Criteria Framework for Security Assessment of Medical Device Software.** International Journal of Intelligent Engineering and Systems, vol. 13, pp. 51-62, 2020. DOI: 10.22266/ijies2020.1031.06

LANGUAGES

Arabic - Native • English - Professional Proficiency